



Original Dataset



Question q_i :

Which feature named for he Scottish Gaelic word for lake is south of Beinn Alligin?



Source Context C_i

Wikipedia passages and related documents ...



Supporting Facts S_i

Fact 1 Loch is the Irish, Scottish Gaelic and Scots word for a lake or a sea inlet.

Fact 2 Beinn Alligin one of the classic mountains ...



Gold Answer y_i^*

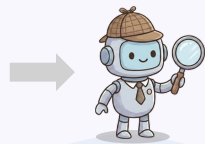
Loch Torridon

Adaptive Fact-Restatement Training Data Construction



Teacher-Guided Initial Adaptive Trace Generation

Input: (q_i, C_i, S_i)



Teacher Reasoning Model $\mathcal{M}_R$

I can decide:



When and where to decide



Granularity and Form of Citation

Generate

Adaptive Reasoning Trace:

I first identify the Scottish Gaelic word for "lake". The text states

<evidence 1> Loch is the Irish, Scottish Gaelic ... **</evidence>**, so ...

Next, I locate Beinn Alligin: **<evidence 2>** Beinn Alligin ... lying to the north of Loch Torridon **</evidence>**, which means Loch Torridon lies to its south.

Therefore, the feature south of Beinn Alligin named for the Gaelic word for lake is **Loch Torridon**.



High-Quality Trace Filtering



Format R_{fmt}



Traceability R_{trc}



Sufficiency R_{suf}



Accuracy R_{acc}

Filter

$\mathcal{D}_{training}$



Length-Diverse Reasoning Trajectory Construction



Original Context Paragraph (C_i)



Distractor Paragraph

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Beinn Alligin ... lying to

If you would like to ...

Length Bucketing

